

Lab Assignment 4

CS 331: Machine Learning

Instructor: Dr. Divya Padmanabhan, Dr. Satyanath Bhat

Teaching Assistants: Vipin, Tina, Manvi, Nimra, Sunny

Feb 3, 2025. To be done individually

1. Attached is the Old Faithful dataset. The dataset consists of 272 measurements of the eruption of the Old Faithful geyser at Yellowstone National Park in the USA. Each row provides a measurement of the duration of an eruption and the time to the next eruption, all in minutes.

- (a) In your code first load the dataset and standardize the data to have zero mean and standard deviation of 1 for each feature, this is to avoid domination of large valued features. This can be done by transforming each point as follows:

$$x_i^{\text{new}} = \frac{x_i - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation values of feature i respectively, x_i represents feature i of a point x .

- (b) With $K = 2$ and initial centers as $[-1, 1]$ and $[1, -1]$ write code for performing K-means on the standardized dataset. K-means algorithm tries to minimize the distance of each point to its assigned center. Let $J = \sum_{i=1}^n \sum_{j=1}^K r_{ij} \|\mathbf{x}_i - \boldsymbol{\mu}_j\|^2$ where $r_{ij} = 1$ if point $\mathbf{x}_i$ is assigned to cluster j , $\boldsymbol{\mu}_j$ refers to center of cluster j , $n = 272$. Compute the value of J in each iteration in your code.
2. Provide plots showing how the cluster means evolve per iteration [similar to Figure 9.1 in Christopher Bishop, Pattern Recognition].
 3. Provide a plot of J in each iteration [similar to Figure 9.2 in Christopher Bishop, Pattern Recognition].

You need to submit your code as well as all the plots on classroom.